

Customer Behavior Analytics and Churn Prediction for E-Commerce Platforms

National Service Application Assignment

Position Applied For: Intelligent Systems and Security Officer (Data Science & Analytics)

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Executive Summary

Customer retention is a critical challenge for e-commerce businesses. This project analyzes customer behavior data to identify patterns that influence customer churn and develops a machine learning model capable of predicting customers who are likely to discontinue their engagement with the platform.

Using an e-commerce customer behavior dataset containing 350 customer records, exploratory data analysis, feature engineering, and predictive modeling techniques were applied. A Random Forest Classifier was developed to predict customer churn based on demographic, transactional, and behavioral factors.

The analysis revealed that customer inactivity, satisfaction level, discount utilization, and customer value are the most influential factors affecting churn. The final model achieved an accuracy of 98.57%, demonstrating strong predictive performance.

1. Introduction

In today's highly competitive e-commerce environment, customer retention has become a critical factor in achieving sustainable business growth. Retaining existing customers is often more cost-effective than acquiring new ones, making it essential for organizations to understand customer behavior and identify factors that influence customer loyalty and engagement.

Customer churn, which refers to customers discontinuing their relationship with a business, can result in significant revenue loss and reduced market competitiveness. As a result, businesses increasingly rely on data analytics and machine learning techniques to predict customer churn and implement proactive retention strategies before customers leave.

The ability to accurately predict churn provides substantial business value. It enables organizations to identify at-risk customers, optimize marketing campaigns, improve customer satisfaction, and allocate resources more effectively. By understanding the characteristics and behaviors associated with churn, businesses can develop targeted interventions that enhance customer retention and long-term profitability.

The objective of this project is to analyze customer behavior data from an e-commerce platform, identify key factors influencing customer churn, and develop a predictive machine learning model capable of classifying customers based on their likelihood of churn. Additionally, the project aims to generate actionable business insights and recommendations that can support data-driven decision-making and improve customer retention outcomes.

2. Dataset Description

The dataset used for this project was sourced from Kaggle and contains customer demographic, transactional, and behavioral information from an e-commerce environment. The dataset was designed to support customer analytics, purchasing behavior analysis, and churn prediction, making it suitable for developing data-driven retention strategies.

Dataset Characteristics

The dataset consists of **350 customer records** and **11 features**, capturing key aspects of customer demographics, purchasing activity, engagement levels, and satisfaction. These variables provide valuable insights into customer behavior and enable the identification of factors that contribute to customer retention and churn.

Key Variables

The dataset includes the following attributes:

- **Customer ID** – Unique identifier assigned to each customer.
- **Gender** – Customer gender classification.
- **Age** – Age of the customer.
- **City** – Customer location.
- **Membership Type** – Customer membership category within the platform.
- **Total Spend** – Total amount spent by the customer.
- **Items Purchased** – Number of items purchased.
- **Average Rating** – Average rating provided by the customer.
- **Discount Applied** – Indicates whether discounts were utilized during purchases.
- **Days Since Last Purchase** – Number of days elapsed since the customer's most recent purchase.
- **Satisfaction Level** – Overall customer satisfaction rating.

Target Variable Definition

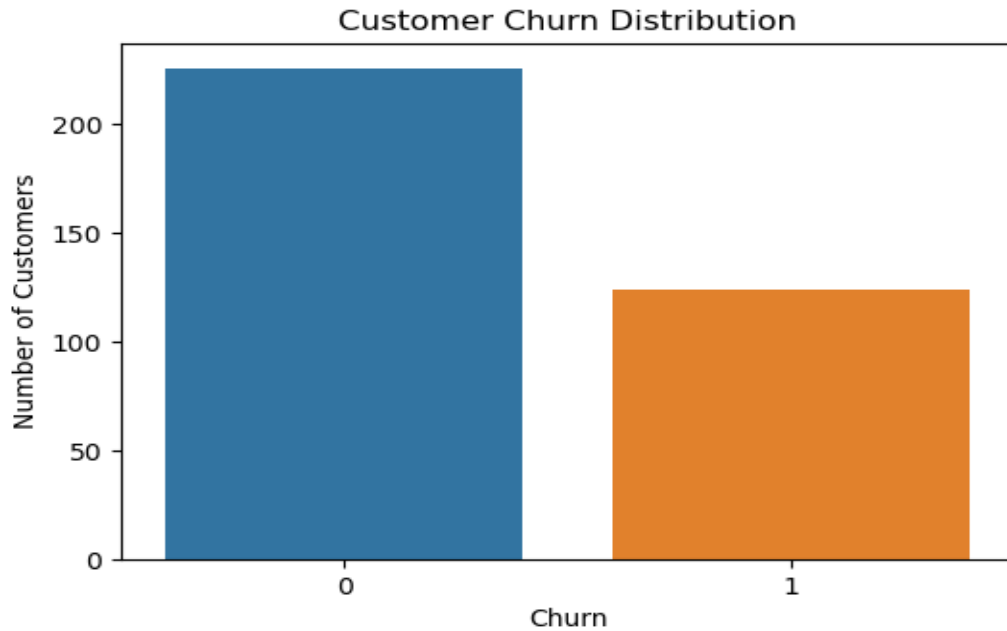
Since the dataset did not contain a predefined churn indicator, a target variable named **Churn** was engineered to facilitate predictive modeling.

Customers were classified as **churned** if they had not made a purchase within the last **30 days**, while customers who made a purchase within 30 days were classified as **active**.

The churn variable was defined as follows:

- **Churn = 1:** Days Since Last Purchase > 30
- **Churn = 0:** Days Since Last Purchase ≤ 30

Figure 1: Distribution of active and churned customers in the dataset.



This business rule was adopted to simulate a realistic customer retention scenario and provide a basis for developing a machine learning model capable of identifying customers at risk of churn.

3. Methodology

This project adopted a structured data analytics and machine learning workflow to analyze customer behavior and develop a predictive churn model. The methodology consisted of four key phases: data preparation, exploratory analysis, feature engineering, and predictive modeling.

3.1 Data Cleaning and Preparation

The dataset was first examined to ensure data quality and consistency. Missing values, duplicate records, and data integrity issues were identified and addressed before proceeding with the analysis.

Key data preparation activities included:

- Identifying and assessing missing values across all variables.
- Replacing missing values in the **Satisfaction Level** attribute using the mode of the column.
- Verifying the dataset for duplicate records and inconsistencies.
- Creating a target variable (**Churn**) based on customer inactivity patterns.

These preprocessing steps ensured that the dataset was suitable for reliable analysis and model development.

3.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to gain insights into customer demographics, purchasing behavior, and churn characteristics. Various statistical summaries and visualizations were used to identify patterns, trends, and relationships within the data.

The analysis focused on:

- Customer demographic distribution.
- Membership type distribution.
- Spending behavior and purchasing patterns.
- Customer satisfaction levels.
- Churn distribution and retention trends.
- Relationships between customer attributes and churn.

The insights obtained from this phase informed the feature engineering and modeling stages.

3.3 Feature Engineering

To enhance the predictive capability of the machine learning model, additional business-oriented features were created from the existing variables. These engineered features were designed to capture customer value, engagement, and purchasing behavior more effectively.

The engineered features included:

- **Customer Lifetime Value (CLV):** Estimated customer value based on total spending and purchase volume.
- **Engagement Score:** A measure of customer engagement derived from purchase activity and ratings.
- **High-Value Customer Indicator:** Classification of customers based on spending behavior.
- **Spending Category:** Segmentation of customers into spending tiers (Low, Medium, and High).

These features provided additional context that improved the model's ability to identify customers at risk of churn.

3.4 Predictive Modeling

A **Random Forest Classifier** was selected as the machine learning algorithm for churn prediction due to its robustness, ability to handle mixed data types, and effectiveness in classification tasks.

Prior to model training, categorical variables were encoded into numerical representations suitable for machine learning algorithms. The dataset was then divided into training and testing subsets using an **80:20 split**, where:

- **80%** of the data was used for model training.
- **20%** of the data was reserved for model evaluation.

The trained model was evaluated using classification metrics, including accuracy, precision, recall, and F1-score, to assess its predictive performance and reliability.

4. Results and Findings

4.1 Model Performance

To predict customer churn, a Random Forest Classifier was developed using customer demographic, transactional, and behavioral data. The model was evaluated on a test dataset comprising 20% of the total observations.

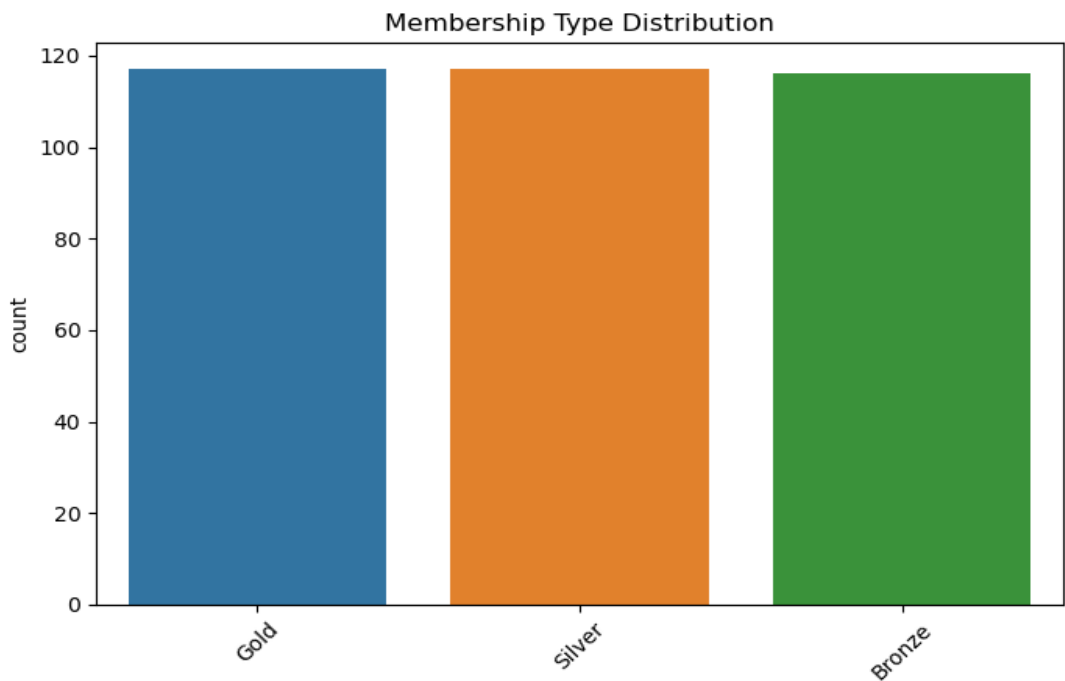
Model Evaluation Results

Metric Score

Accuracy 98.57%

The model demonstrated excellent predictive performance, correctly classifying the vast majority of customers in the test dataset. The high accuracy score indicates that the selected features effectively captured the patterns associated with customer churn behavior.

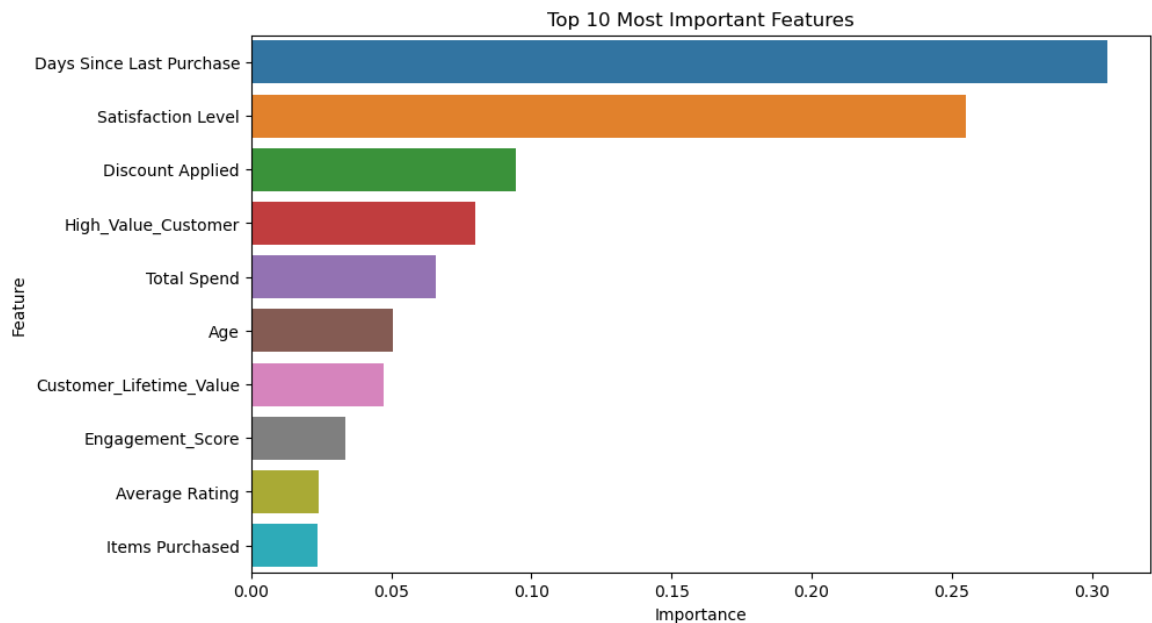
Figure 2: Distribution of churn across different membership categories.



Feature Importance Analysis

Feature importance analysis was conducted to identify the variables that contributed most significantly to churn prediction.

Figure 3: Top predictors influencing customer churn.



The analysis revealed that customer inactivity, satisfaction level, discount utilization, and customer value metrics were among the most influential predictors of churn.

4.2 Key Findings

Finding 1: Customer Inactivity is the Strongest Indicator of Churn

The analysis identified **Days Since Last Purchase** as the most important predictor of customer churn. Customers who had not made purchases for extended periods were significantly more likely to discontinue their engagement with the platform.

This finding highlights the importance of monitoring customer activity levels and implementing timely re-engagement strategies to reduce customer attrition.

Finding 2: Customer Satisfaction Significantly Influences Retention

Customer satisfaction emerged as the second most influential factor affecting churn. Customers reporting lower satisfaction levels were more likely to disengage from the platform compared to highly satisfied customers.

This emphasizes the need for organizations to continuously monitor customer experiences and address service issues proactively.

Finding 3: Discount Usage Impacts Customer Loyalty

The analysis revealed that the application of discounts plays a meaningful role in customer retention. Customers who benefited from promotional offers and discounts demonstrated different retention patterns compared to those who did not.

This suggests that targeted promotional campaigns can serve as an effective mechanism for reducing churn among at-risk customers.

Finding 4: High-Value Customers Exhibit Distinct Purchasing Behaviors

Customers classified as high-value customers based on their spending behavior showed unique purchasing and engagement patterns. These customers contribute significantly to overall business revenue and therefore represent a critical segment for retention efforts.

Organizations can benefit from implementing personalized loyalty programs and premium customer engagement initiatives aimed at retaining these valuable customers.

4.3 Summary of Findings

The results indicate that customer churn is influenced by a combination of behavioral, transactional, and satisfaction-related factors. In particular, customer inactivity and satisfaction levels emerged as the strongest indicators of churn risk.

The predictive model achieved strong performance and demonstrated the effectiveness of data-driven approaches in identifying customers at risk of leaving. These insights can support proactive retention strategies, improve customer experience, and ultimately contribute to increased business profitability.

5. Business Recommendations

Based on the findings from the analysis and predictive modeling process, the following recommendations are proposed to improve customer retention and reduce churn within the e-commerce platform.

Recommendation 1: Implement Automated Customer Re-Engagement Campaigns

Since customer inactivity was identified as the strongest predictor of churn, the organization should establish automated re-engagement campaigns targeting customers who have not made purchases within a specified period. Personalized emails, reminders, and promotional offers can be used to encourage customers to return before they become inactive for extended periods.

Recommendation 2: Strengthen Customer Satisfaction Monitoring

Customer satisfaction was found to have a significant impact on retention. The organization should implement continuous customer feedback mechanisms, including surveys, ratings, and customer support evaluations, to identify dissatisfaction early and address potential issues before they lead to customer attrition.

Recommendation 3: Develop Loyalty and VIP Retention Programs

High-value customers contribute substantially to overall business performance and revenue generation. To retain these valuable customers, the organization should introduce loyalty programs, exclusive rewards, personalized recommendations, and premium customer experiences that encourage long-term engagement and repeat purchases.

Recommendation 4: Utilize Targeted Promotional Strategies

The analysis demonstrated that discounts and promotional incentives influence customer retention behavior. Rather than applying broad discounts across all customers, the organization should leverage predictive analytics to identify customers at risk of churn and provide targeted offers designed to encourage continued engagement and purchasing activity.

6. Conclusion

This project successfully applied data analytics and machine learning techniques to analyze customer behavior and predict customer churn within an e-commerce environment. Through data cleaning, exploratory data analysis, feature engineering, and predictive modeling, valuable insights were obtained regarding the factors that influence customer retention.

The Random Forest Classifier achieved an accuracy of **98.57%**, demonstrating strong predictive capability in identifying customers at risk of churn. The analysis revealed that customer inactivity, satisfaction level, discount utilization, and customer value are among the most influential factors affecting retention outcomes.

The findings highlight the value of data-driven decision-making in improving customer retention strategies and enhancing business performance. By implementing proactive retention initiatives based on the insights generated from this analysis, organizations can strengthen customer relationships, reduce churn, and improve long-term profitability.

Overall, this project demonstrates how data science and analytics can be effectively leveraged to transform customer data into actionable business intelligence, supporting strategic decision-making and sustainable organizational growth.